

3.4 Addendum: bias correction in SA algorithms

ESTP Training
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Description of the problem

- Log-additive decompositions
 - SEATS with log transformation, X11 with log-additive mode

$$\log Y_t = t_t + s_t + i_t$$

$$\rightarrow Y_t = \exp t_t \cdot \exp s_t \cdot \exp i_t = T_t \cdot S_t \cdot I_t$$

$$\sum_{\text{yearly}} s_t \equiv 0 \leftrightarrow \sqrt[p]{\exp s_t} \equiv 1 \leftrightarrow \frac{1}{p} \sum_{\text{yearly}} S_t > 1$$

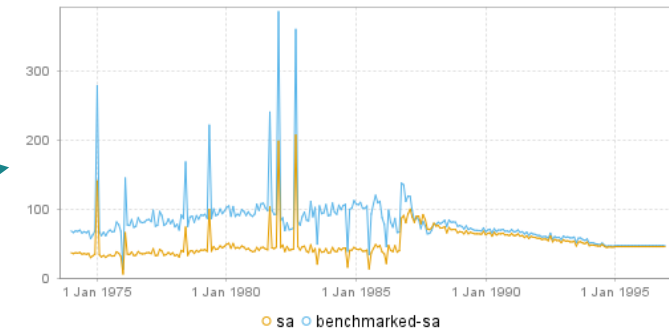
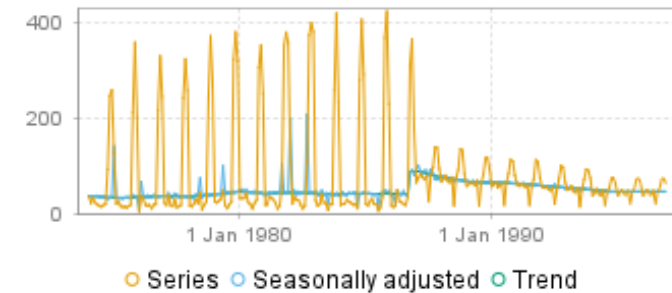
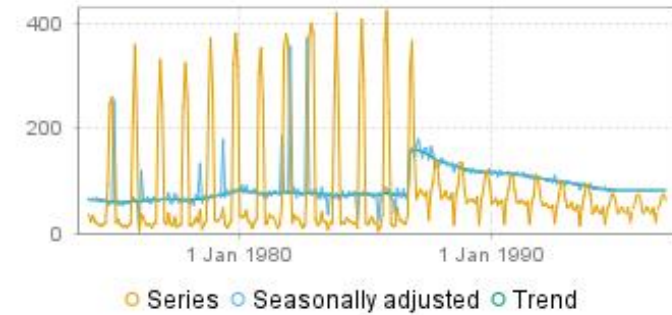
(the arithmetic mean is always higher than the geometric mean)

$$\frac{1}{p} \sum_{\text{yearly}} SA_t < \frac{1}{p} \sum_{\text{yearly}} Y_t$$



Possible solutions

- Avoid log transformations (multiplicative X11)
- Correct S, I and T for their average bias on the whole series (Tramo)
 - But...
- Don't correct the bias
- Experimental options (X12): smooth correction
- Don't correct and apply benchmarking



Bias co



Additional remarks

- Bias correction will generate some revisions (constant factor)
- Currently no good solution in JD+ if the seasonal factors highly differ following the periods



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